BRNO UNIVERSITY OF TECHNOLOGY
VYSOKÉ UČENÍ TECHNICKÉ V BRNĚ

FACULTY OF INFORMATION TECHNOLOGY
FAKULTA INFORMAČNÍCH TECHNOLOGIÍ

DEPARTMENT OF COMPUTER SYSTEMS
ÚSTAV POČÍTAČOVÝCH SYSTÉMŮ

3D NEURAL CELLULAR AUTOMATA

3D NEURÁLNÍ CELULÁRNÍ AUTOMATY

BACHELOR'S THESIS
BAKALÁŘSKÁ PRÁCE

AUTHOR **MICHAL REPČÍK**
AUTOR PRÁCE

SUPERVISOR **Ing. KAREL FRITZ,**
VEDOUCÍ PRÁCE

BRNO 2026

Abstract

Do tohoto odstavce bude zapsán výtah (abstrakt) práce v anglickém jazyce.

Abstrakt

Do tohoto odstavce bude zapsán výtah (abstrakt) práce v českém (slovenském) jazyce.

Keywords

Sem budou zapsána jednotlivá klíčová slova v anglickém jazyce, oddělená čárkami.

Klíčová slova

Sem budou zapsána jednotlivá klíčová slova v českém (slovenském) jazyce, oddělená čárkami.

Reference

REPČÍK, Michal. *3D Neural Cellular Automata*. Brno, 2026. Bachelor's thesis. Brno University of Technology, Faculty of Information Technology. Supervisor Ing. Karel Fritz,

3D Neural Cellular Automata

Declaration

I hereby declare that this Bachelor's thesis was prepared as an original work by the author under the supervision of Ing. Karel Fritz I have listed all the literary sources, publications and other sources, which were used during the preparation of this thesis. I have used ChatGPT and Grammarly to refine the text and correct spelling and other language mistakes. I used Copilot when working on the software.

. . . . . .
Michal Repčík
March 29, 2026

Acknowledgements

I would like to express my sincere thanks to my supervisor, Ing. Karel Fritz, for his guidance and support.

Contents

1 Introduction **3**

2 Abstract **4**

3 Theoretical Foundations **5**
3.1 Prerequisites 5
3.2 Cellular Automata 5
3.3 Neural Cellular Automata 8
3.4 Extension from Two to Three Dimensions 8
3.5 Core Components of a 3D Neural Cellular Automaton 9

4 Related Work **13**
4.1 Growing Neural Cellular Automata 13
4.2 Three-Dimensional Neural Cellular Automata in Interactive Environments . 16
4.3 Open-Source and Applied Three-Dimensional Neural Cellular Automata . . 17
4.4 Chapter Conclusion 18

5 Architectural Design **19**
5.1 Design Goals and Requirements 19
5.2 System Overview 20
5.3 Core NCA Model 20
5.4 Training Pipeline 23
5.5 Server and Networking 24
5.6 Core Framework 24
5.7 NCA Model Components 24
5.8 Training Pipeline 24
5.9 Server 25
5.10 Blender Add-on 25

6 Testing **26**

7 Experiments **27**

8 Conclusion **28**

Bibliography **29**

List of Figures

5.1 High-level system architecture. Arrows indicate dependency direction. The dashed TCP boundary marks the network separation between the server and the Blender client. 20

5.2 Per-voxel state vector layout. The vector contains hidden channels, optional task channels encoded as a one-hot vector, and visible channels. The alpha channel, which serves as the alive signal, is always the last visible channel. . 21

5.3 Data flow through a single NCA update step. The state passes through the fixed 3D perception filters, is transformed by the shared update rule, and is then gated by the alive masking mechanism before producing the updated state. 22

5.4 Detailed data flow through a single NCA update step, showing the full layer sequence within the update MLP, the optional stochastic fire mask, and the bounded tanh output. 24

5.5 Wire message format. Each message consists of a four-byte big-endian unsigned integer carrying the payload byte length, followed by the UTF-8 JSON payload. 25

5.6 Protocol message types and their JSON fields. Messages above the divider are sent by the client; messages below are sent by the server. 25

Chapter 1

Introduction

TODO

Chapter 2

Abstract

TODO

Chapter 3

Theoretical Foundations

3.1 Prerequisites

This work assumes familiarity with convolutional neural networks, gradient-based optimisation, and automatic differentiation as implemented in frameworks such as PyTorch [17, 3]. Goodfellow et al. [6] and the official PyTorch documentation [3] provide comprehensive introductions to these topics.

The remainder of this chapter is self-contained with respect to cellular automata and their neural extensions.

3.2 Cellular Automata

Cellular automata (CA) are discrete dynamical systems composed of a regular grid of cells, each occupying one state from a finite set. All cells update in parallel at each discrete time step according to a local rule that depends only on the states of neighbouring cells [12].

A cellular automaton is defined by:

- a **lattice** of cells, typically arranged in one, two, or three dimensions,
- a **finite state set** $\mathcal{S}$,
- a **neighbourhood function** $\mathcal{N}$ describing which cells influence a given cell,
- an **update rule** $f : \mathcal{S}^{|\mathcal{N}|} \to \mathcal{S}$ that determines the next state [8].

Ulam and von Neumann developed the first formal CA models in the 1940s as a theoretical framework for self-replicating machines [15]. Von Neumann's original construction used a two-dimensional lattice with 29 distinct cell states and demonstrated that a sufficiently complex local rule can produce a system capable of constructing copies of itself. This result established that self-replication does not require a centralised blueprint but it can emerge from purely local interactions on a discrete grid.

Wolfram later proposed a behavioural classification of CA into four qualitative classes based on their long-term dynamics [24, 25]. Class 1 systems converge to a uniform state regardless of initial conditions. Class 2 systems settle into static or periodic patterns. Class 3 systems produce chaotic, seemingly random output sensitive to initial conditions. Class 4 systems generate complex dynamics with long-lived interacting structures capable of computation. This classification provides a useful reference for characterising the dynamics of both classical and neural CA.

3.2.1 One-Dimensional Cellular Automata

The simplest non-trivial CA operate on a one-dimensional array of cells. In the elementary CA family, each cell reads its own state and its two immediate neighbours (radius 1, neighbourhood size 3). With binary states, $2^3 = 8$ neighbourhood configurations are possible, giving $2^8 = 256$ distinct rules in Wolfram's enumeration [24].

Each rule number encodes a next-state lookup table in binary. For example, Rule 90 assigns the next state of a cell based on the exclusive-or (XOR) of its two neighbours, producing the Sierpiński triangle from a single active seed. This pattern is fractal and self-similar, placing Rule 90 in Wolfram's Class 3 (chaotic behaviour). Rule 30 also belongs to Class 3 and produces irregular, aperiodic evolution that has been used as a source of pseudorandomness [25].

Rule 110 (Class 4) is the smallest known Turing-complete one-dimensional CA. Cook [4] proved that it supports gliders and glider collisions that can simulate any Turing machine. This result was significant because it showed that computational universality can arise from an extremely simple local rule applied to a one-dimensional lattice.

Despite their simplicity, elementary CA demonstrate all four classes of Wolfram's classification. They provide a minimal setting for studying how local rules produce global behaviour, a question that extends directly to the neural CA models discussed later in this chapter.

3.2.2 Two-Dimensional Cellular Automata

In two dimensions, the Moore neighbourhood (3×3 block surrounding each cell) is the most common choice. With binary states and a Moore neighbourhood, the update rule must handle $2^9 = 512$ possible input configurations. Because the full lookup table is large, 2D rules are typically expressed using birth/survival notation. A rule written as Bn/Sm specifies that a dead cell becomes alive (is "born") when it has exactly n live neighbours, and a living cell survives when it has exactly m live neighbours. All other cells die or remain dead.

Conway's Game of Life [5] uses the rule B3/S23: a dead cell becomes alive with exactly three live neighbours, and a living cell survives with two or three live neighbours. Despite the simplicity of this rule, Life produces a wide variety of emergent structures. Still lifes (such as the block and beehive) are stable configurations that do not change between steps. Oscillators (such as the blinker and toad) cycle through a fixed sequence of states. Spaceships (such as the glider and lightweight spaceship) translate across the grid while cycling through their phases. More complex constructions include glider guns, which periodically emit spaceships, and computational structures that perform logic operations.

Berlekamp, Conway, and Guy [1] showed that Life is Turing-complete: any computation that can be expressed as a Turing machine can, in principle, be carried out within a sufficiently large Life grid. This result highlights a key property of cellular automata — complex, globally coordinated behaviour can emerge from a homogeneous local rule. The same property later motivated the development of neural cellular automata, where the local rule is learned rather than hand-designed.

3.2.3 Three-Dimensional Cellular Automata

In three dimensions, the Moore neighbourhood covers 26 adjacent cells (6 face-adjacent, 12 edge-adjacent, and 8 corner-adjacent) in a $3 \times 3 \times 3$ cube surrounding the central cell.

Binary states with a full Moore neighbourhood produce $2^{27} \approx 134$ million neighbourhood configurations, making exhaustive rule enumeration impractical. For this reason, practical 3D CA use totalistic or outer-totalistic rules that depend only on the count of live neighbours rather than their exact positions [25].

Several notable 3D CA have been studied. Von Neumann's 29-state self-replicating automaton [15], originally formulated in 2D, has been extended to three dimensions, where the additional spatial freedom allows more compact replicating structures. Excitable-media rules such as Brian's Brain have also been generalised to 3D lattices, producing travelling wavefronts and spiral structures in three-dimensional space [10]. Three-dimensional Game of Life variants using totalistic birth/survival rules generate branching and crystal-like growth patterns that have no direct 2D analogue.

Hardware advances in the 2000s made large-scale 3D CA simulations practical on consumer hardware [22]. GPU-accelerated implementations can now run volumetric CA with grid sizes exceeding 128^3 cells in real time. These volumetric models provided a natural setting for the neural CA extensions discussed later in this chapter, as the underlying lattice structure and local-update paradigm transfer directly from classical to neural formulations.

3.2.4 Differentiability and Continuous Cellular Automata

Classical CA use discrete states and discrete update rules. However, several extensions have shown that the core CA principles (spatial lattice, local interactions, and parallel updates) also work with continuous states and smooth transition functions. These continuous CA are directly relevant to neural CA, which operate on real-valued state vectors and use differentiable update rules trained by gradient descent.

An early theoretical connection was established by Turing [23], whose reaction-diffusion model of morphogenesis describes how spatial patterns can arise from the interaction of locally diffusing chemical substances. Although Turing's model uses partial differential equations rather than a discrete lattice, the underlying principle - local interactions producing global spatial structure - is the same principle that governs cellular automata. This connection between reaction-diffusion systems and CA has motivated much of the subsequent work on continuous and neural CA.

SmoothLife [18] generalises Conway's Game of Life to continuous space and time. It replaces the discrete neighbour count with continuous averaging kernels and the binary birth/survival rule with smooth sigmoid transition functions. The result is a system that runs on a continuous 2D field and produces gliders, oscillators, and wave-like structures that closely resemble their discrete counterparts but evolve smoothly in time. SmoothLife demonstrates that the characteristic behaviours of Life do not depend on discretisation, but rather arise from the structure of local interactions.

Lenia [2] extends this approach further by parameterising both the convolution kernel (which defines the neighbourhood shape and weighting) and the growth function (which determines how the local average maps to a state change). Systematic exploration of Lenia's parameter space has produced hundreds of stable, self-organising patterns that resemble biological microorganisms such as orbiting particles, pulsating organisms, and multi-species ecologies. Lenia shows that a rich space of emergent behaviours can be accessed by varying the parameters of a continuous local update rule.

Both SmoothLife and Lenia demonstrate that complex self-organisation does not require discrete state transitions. Local, differentiable interactions are sufficient to produce stable spatial structures. Neural cellular automata build on this observation. Instead of manually

exploring the parameter space as in Lenia, neural CA use gradient-based optimisation to find update rules that produce a specific target pattern. The differentiability of the update rule is what makes this optimisation possible.

3.3 Neural Cellular Automata

Mordvintsev et al. [13] introduced Neural Cellular Automata (NCA) as a differentiable generalisation of classical CA. The discrete lookup-table rule is replaced by a small neural network whose parameters are shared across all cells. The spatial lattice stays discrete, but each cell's state becomes a real-valued vector, and the update rule is trained end-to-end by gradient descent.

Let the lattice be $\Omega \subset \mathbb{Z}^d$, where d is the spatial dimension. Each cell $x \in \Omega$ holds a state vector

$$\mathbf{s}_t(x) \in \mathbb{R}^C,$$

where C is the number of channels. The full grid state at time t is a tensor over all cell states.

The update process has two stages [13]. First, a fixed perception operator extracts local neighbourhood information through convolutional filters. The filters are applied to the full multi-channel grid and produce a perception vector at each cell location. Second, a shared neural network maps this perception vector to a state increment $\Delta\mathbf{s}$ that is added to the current state:

$$\mathbf{s}_{t+1}(x) = \mathbf{s}_t(x) + \Delta\mathbf{s}(x).$$

This residual update formulation keeps the defining CA property - strict locality - while replacing the hand-crafted transition function with a learnable model. Because the update rule is differentiable with respect to its parameters, the automaton can be unrolled over multiple time steps and trained by gradient descent to minimise a loss defined on the resulting configuration.

The key conceptual shift compared to classical and continuous CA is that the update rule is not designed or discovered by exploration, but optimised for a specific objective. The system learns to grow a target pattern from a seed, maintain it over time, or regenerate it after damage depending on the training regime. This makes neural CA a form of learned self-organisation: the local rule is shaped by optimisation, but the global behaviour emerges from distributed local interactions, just as in classical CA.

3.4 Extension from Two to Three Dimensions

The original NCA of Mordvintsev et al. [13] operates on a 2D grid. Extending it to three-dimensional voxel lattices is conceptually straightforward but introduces several practical differences that affect both the architecture and the training procedure.

In 2D, perception uses 3×3 filters over a nine-cell Moore neighbourhood. In 3D, the same filters generalise to $3 \times 3 \times 3$ volumes covering twenty-seven cells. Prior 3D extensions [21, 20, 9] have shown that Sobel-type gradient filters and other perception operators transfer directly to x, y, and z axes without changing the overall update formulation.

However, volumetric grids introduce two important differences. First, cell count grows cubically with resolution. A 32^3 grid contains 32 768 cells; a 64^3 grid contains 262 144. This cubic scaling increases both memory consumption and computational cost per step,

particularly during training when gradients must be stored for backpropagation through multiple steps.

Second, information still propagates only through local interactions, one step at a time. In a 32^3 grid, a signal originating at one corner must traverse roughly 55 steps (the grid diagonal in Manhattan distance through the $3 \times 3 \times 3$ neighbourhood) to reach the opposite corner. In larger grids, this propagation distance grows further. Maintaining coherent global structure therefore requires stable dynamics over longer unrolling horizons than in comparable 2D systems.

As a consequence, behaviours such as persistence and regeneration may behave differently in volumetric domains. The change in neighbourhood structure (27 neighbours instead of 9), spatial topology, and information propagation distance can all affect the stability and character of the emergent dynamics.

3.5 Core Components of a 3D Neural Cellular Automaton

This section describes the principal components of a three-dimensional neural cellular automaton at the conceptual level. The goal is to establish the general design principles that underlie these systems before addressing the specific architecture and implementation in later chapters.

3.5.1 State Representation

Each voxel in a 3D neural CA maintains a vector of C real-valued channels:

$$\mathbf{s} \in \mathbb{R}^C.$$

The channels divide into two functional groups. **Visible channels** represent quantities compared directly to a target pattern, such as colour components (RGB), occupancy, or material properties. **Hidden channels** provide internal memory that is not directly supervised but carries information between neighbouring cells across time steps [13].

The separation between visible and hidden channels is a design choice inherited from the 2D NCA formulation. Visible channels define the observable output of the system, while hidden channels give each cell additional representational capacity to encode information about its local context, history, or role within the emerging structure. Because updates are strictly local, a cell cannot directly observe distant parts of the grid. Hidden channels allow cells to accumulate and relay information over multiple steps, enabling the kind of long-range coordination needed to produce coherent global patterns from local rules.

Most NCA models also designate one channel as a continuous *alive* signal. This channel determines whether a cell is considered active and whether its updates are applied. The alive mechanism appears in the original NCA [13] and has parallels in continuous CA such as SmoothLife [18] and Lenia [2], where real-valued fields govern the activation and deactivation of spatial regions.

3.5.2 Perception Mechanism

Neural CA preserve the locality constraint that defines classical CA: each cell updates based only on its immediate neighbourhood [8, 25]. In a 3D neural CA, locality is realised by applying a fixed set of convolutional filters with kernel size $3 \times 3 \times 3$ to the multi-channel

state grid. Each filter reads the 27-cell neighbourhood of every voxel and produces a scalar output per channel [13, 21].

Common filter types include:

- **Identity filters**, which return the cell's own state unchanged. These provide the update rule with information about what the cell currently contains.
- **Neighbour aggregation filters**, which compute a sum or average of adjacent voxels. These capture the local density and composition of the neighbourhood.
- **Gradient filters**, such as Sobel-type kernels extended to the x, y, and z axes. These approximate directional differences and provide a local analogue of gradient sensing in biological morphogenesis [13, 21, 23].

The outputs of all filters applied across all channels are concatenated into a single perception vector per cell. If the state has C channels and G filter groups are used, the perception vector has dimensionality $G{\cdot}C$. This vector is then passed to the update function as input.

The filters are fixed and non-learnable. Using fixed filters constrains the spatial information available to the update rule, which stabilises training and reduces the number of parameters that must be optimised. The approach follows the convention established by Mordvintsev et al. [13] and adopted in subsequent work [21, 20, 9].

3.5.3 Update Rule

The update rule is a shared parametric function that governs how each cell's state evolves over time. It takes the perception vector as input and produces a state increment $\Delta\mathbf{s}$ added to the current state:

$$\mathbf{s}_{t+1} = \mathbf{s}_t + \Delta\mathbf{s}.$$

This residual formulation encourages incremental modifications rather than abrupt state changes, contributing to dynamical stability over long simulation horizons [13].

The update function is typically implemented as a small multilayer perceptron (MLP) applied independently at each cell. This point-wise application ensures that the update depends only on the locally perceived information, not on any global state. The parameter count is kept deliberately low following the original 2D model, approximately 8 000 trainable parameters [13]. This small scale underlines that the complexity of the resulting structures arises from iterative local interaction, not from network capacity.

Two stabilisation mechanisms appear in most NCA formulations:

- **Alive gating** conditions updates on the alive channel. If a cell and all of its neighbours have alive values below a threshold, the cell's update is set to zero. This prevents empty regions from spontaneously generating structure and provides a soft boundary between active and inactive areas of the grid [13].
- **Stochastic update masking** applies updates to a random subset of cells at each step rather than to all cells simultaneously. This breaks temporal synchronisation artefacts and encourages robustness to asynchronous perturbations [13].

Both mechanisms improve long-term stability and regenerative capacity. They become increasingly important as systems scale to larger grids or higher dimensions, where dynamical instabilities can accumulate over more cells and longer simulation horizons.

3.5.4 Grid Management

The neural CA runs on a finite three-dimensional lattice with fixed dimensions. Boundary conditions (periodic, zero-padded, or reflective) are chosen per experiment and affect the dynamics near the grid edges [13]. Sudhakaran et al. [21] note that the choice of boundary handling varies across implementations and can influence the shapes that the automaton produces.

The primary challenge of 3D grids is their cubic memory scaling. A grid of size N^3 with C channels per cell and 32-bit floating-point precision requires $N^3 \times C \times 4$ bytes of storage per state. For a 32^3 grid with 20 channels, this is approximately 2.5 MB per state. During training, multiple states are held simultaneously (for batching and pool-based sampling), and gradient storage can multiply memory requirements further.

Research on large-scale volumetric simulations addresses this through specialised data structures. Sparse Voxel Octrees [11] exploit the observation that most voxels in a large grid are empty, storing only occupied regions in a hierarchical tree. OpenVDB [14] provides a similar approach with dynamic topology, allowing efficient insertion and deletion of voxels during simulation. While the framework developed in this work uses dense grids at moderate resolution, these sparse representations offer a path toward scaling neural CA to larger volumetric domains in future work.

Initialisation is another practical consideration. Most neural CA start from a *seed* – a single active cell or small cluster placed at the centre of an otherwise empty grid. The alive channel of the seed cell is set to a positive value, and the system must grow the target structure from this minimal starting point through repeated application of the local update rule.

3.5.5 Training and Optimisation

Because the update rule is differentiable, the parameters of a neural CA can be optimised by gradient descent. The standard training procedure unrolls the automaton for T steps starting from a seed state, computes a loss between the resulting configuration and a target, and backpropagates through the full unrolled computation graph [13].

The most common loss function is mean squared error (MSE) between the predicted and target visible channels. For structural objectives, intersection-over-union or other task-specific metrics may be used [16]. The optimiser is typically Adam or one of its variants, with learning rates in the range of 10^{-4} to 10^{-2} [21].

Several training strategies have been developed to encourage specific behaviours:

- **Pool-based sampling** maintains a pool of persistent states at various stages of growth. Each training iteration samples a batch from the pool, runs the automaton forward, computes the loss, and writes the updated states back. This exposes the model to states at different developmental stages rather than always starting from a fresh seed [13].

- **Curriculum learning** gradually increases the number of unrolled steps during training. Early iterations use few steps to learn coarse shape, while later iterations use many steps to refine details and test long-term stability.

- **Training regime selection** determines which aspects of the dynamics are explicitly optimised. These regimes (growth, persistence, regeneration) are discussed in detail in the Related Work chapter.

Backpropagation through many unrolled steps requires storing intermediate activations for the entire sequence, which can exhaust GPU memory. Gradient checkpointing [17] addresses this by discarding intermediate activations during the forward pass and recomputing them during the backward pass, trading approximately twice the computation for substantially reduced memory usage.

Chapter 4

Related Work

This chapter reviews the research most directly related to neural cellular automata and their three-dimensional extensions. The focus is on architectures, training procedures, and experimental results. Each section identifies limitations or open questions that motivate the framework developed in this work.

Several of the works discussed here were already introduced in Chapter 3 to establish formal background. Here, the focus shifts to their concrete design decisions, findings, and the specific choices that inform the approach taken in this work.

4.1 Growing Neural Cellular Automata

Mordvintsev, Randazzo, Niklasson, and Levin [13] introduced the Growing Neural Cellular Automata paradigm that is the methodological basis for this work. They replaced the discrete lookup-table rules of classical CA with a small shared neural network trained end-to-end by gradient descent. The model preserves strict spatial locality while learning to grow and maintain target patterns from simple initial conditions – typically a single live cell.

Unlike earlier continuous CA such as Lenia [2], which rely on manual parameter exploration to discover interesting behaviours, neural CA use supervised optimisation: the transition function is trained to produce a specific target pattern. This work extends that task-driven optimisation approach to three dimensions.

Chapter 3 covered the general NCA principles. The focus here is on the concrete design and empirical properties of the 2020 implementation: its architecture, stabilisation mechanisms, training procedure, and experimental results.

4.1.1 State Representation

Each cell in the 2D model holds a 16-channel real-valued state vector. Four channels encode RGBA colour values compared against a target image. The remaining twelve channels serve as hidden memory.

The hidden channels work as a distributed memory system embedded in the lattice. Because updates are strictly local, long-range coordination must emerge through repeated short-range interactions. A cell cannot observe the state of a distant cell directly; it can only receive information that has been relayed through intermediate cells over multiple steps. The hidden channels provide the capacity for this relay process.

This design is central to the stability and regenerative behaviour observed in later experiments. Reducing the number of hidden channels degrades pattern quality and stability, while increasing it beyond a certain point yields diminishing returns [13]. The balance between enough memory for coordination and a compact state for efficient training directly informs the channel configurations used in subsequent 3D extensions.

4.1.2 Perception Mechanism

Perception uses a fixed set of non-learnable convolutional filters applied to the full multi-channel grid:

- An identity filter, returning each cell's own state,
- Sobel filters approximating spatial derivatives in the x and y directions,
- A Laplacian filter capturing local curvature.

Applied across all 16 channels, these three filter types produce a $3 \times 16 = 48$-dimensional perception vector at each spatial location. The Sobel filters detect directional gradients in the neighbourhood, for example where a pattern transitions from occupied to empty, while the Laplacian identifies edges and boundaries by measuring how much a cell's state differs from its neighbours' average.

A small MLP operating independently at each cell transforms this perception vector into a state increment. The perception step introduces no learnable parameters; all learning occurs in the MLP.

Fixed perception filters are used to ensure locality and translation invariance by design, while also stabilising optimisation by limiting the spatial information available to the update rule. This choice becomes particularly important in volumetric extensions, where the neighbourhood expands from 9 to 27 cells and the perception vector grows correspondingly in dimensionality.

4.1.3 Update Rule and Parameter Sharing

The update rule is a lightweight neural network shared across all cells. It produces a state increment added residually to the current state:

$$\mathbf{s}_{t+1} = \mathbf{s}_t + \Delta\mathbf{s}.$$

Residual updates encourage incremental modifications rather than abrupt transitions, contributing to dynamical stability. Because the same parameters apply at every spatial location, the learned rule defines a homogeneous dynamical system over the entire grid. No cell has a special role by construction, this means that any differentiation in cell behaviour must emerge from the spatial context encoded in the perception vector.

The full architecture has approximately 8 000 trainable parameters. This small scale is deliberate: it shows that the complexity of the resulting structures comes from iterative local interaction, not from model capacity. By contrast, a standard convolutional neural network for image generation may have millions of parameters. The NCA achieves comparable visual complexity through temporal depth (running many update steps) rather than through network width.

Stabilisation Mechanisms

Two mechanisms are critical for robust long-term dynamics:

- **Alive gating** conditions state updates on whether the designated alpha channel exceeds a threshold in the local neighbourhood. This creates a soft alive/dead boundary and prevents inactive regions from destabilising active structures. Without alive gating, empty regions can spontaneously develop non-zero states due to numerical noise, leading to spurious growth outside the intended pattern.
- **Stochastic update masking** applies updates to a random subset of cells at each step. At each time step, each cell independently decides (with some probability, e.g., 0.5) whether to apply its computed update or skip it. This breaks synchronisation artefacts - patterns that form only because all cells update in lockstep – and produces dynamics that are robust to asynchronous perturbations.

Both mechanisms improve long-term stability and regenerative capacity. Horvath and Levin [7] later demonstrated that stochastic masking also encourages the emergence of distinct cell types within the same NCA, as cells must develop local strategies that work regardless of which neighbours updated in the current step.

4.1.4 Training Regimes

Mordvintsev et al. [13] introduced three training regimes: growing, persistent, and regenerating. Each regime targets a different aspect of the learned dynamics. The choice of regime determines the balance between initial pattern formation, long-term stability, and robustness to damage. These regimes define the training methodology adopted and extended in this work.

Growing Regime

The model is trained to produce a target image from a single active seed. Loss is computed after a fixed number of update steps T, and the model learns to expand outward and approximate the target pattern within that horizon.

No constraint is imposed beyond step T. Patterns may drift, decay, or diverge if simulation continues past the trained step count. In practice, growing-only models commonly exhibit instability within 100–200 steps beyond their training horizon [13].

Persistent Regime

The persistent regime evaluates loss at multiple time steps beyond initial convergence. Rather than stopping at step T, training continues to evaluate the model at steps T, $T+\delta$, $T+2\delta$, and so on, penalising any drift away from the target.

This produces more stable dynamics. Mordvintsev et al. observed that persistent models also exhibit limited regenerative behaviour even without explicit damage training, suggesting that the stability mechanisms learned for persistence partially generalise to perturbation recovery.

Randazzo et al. [19] later extended this idea by training NCA with persistent memory, showing that models trained for stability over very long horizons develop increasingly robust internal representations in their hidden channels.

Regenerating Regime

During training, random damage is applied by zeroing out contiguous rectangular regions of the grid. The model must reconstruct the target from the damaged configuration.

This explicit objective produces the most robust dynamics. Regenerating models recover from substantial perturbations and maintain structural integrity over long simulation horizons. The regenerative capacity is qualitatively reminiscent of biological organisms that can regrow damaged tissue from partial structure [13].

4.1.5 Relevance to This Work

This work adopts the architecture, training regimes, and optimisation strategy of Mordvintsev et al. [13] as its baseline and extends the framework to three-dimensional voxel lattices. The transition from 2D to 3D increases neighbourhood size from 9 to 27 cells, increases computational cost cubically, and extends the information propagation distance required for global coordination. Whether the stability and regenerative behaviours observed in planar domains carry over to volumetric grids is a central question motivating the experiments in this work.

4.2 Three-Dimensional Neural Cellular Automata in Interactive Environments

Sudhakaran et al. [21] extended NCA to three dimensions within the sandbox game *Minecraft*, evaluating volumetric growth of structures and functional machines under real-time constraints and user interaction.

Their model extends the 2D growing NCA paradigm to a 3D voxel grid. Each voxel stores a state vector with visible and hidden channels. Perception uses local $3 \times 3 \times 3$ neighbourhoods with identity and gradient features, processed by a small neural network applied identically at every voxel. The system is trained by gradient descent to grow a target structure from a single seed.

The authors showed that neural CA can produce coherent volumetric structures in a dynamic, interactive environment. The learned rules stay local and homogeneous, preserving the CA paradigm while enabling complex global structure formation. Their experiments showed convergence toward target voxel configurations and partial regenerative behaviour under perturbation: when parts of a Minecraft structure were manually destroyed, the NCA could partially rebuild the damaged region.

An interesting aspect of this work is its evaluation in a constrained real-time environment. The Minecraft platform imposes fixed block types, a specific rendering pipeline, and interaction mechanisms that differ from the unconstrained continuous-valued grids used in purely computational experiments. Despite these constraints, the NCA paradigm produced recognisable structures, showing that the approach is not limited to idealised simulation environments.

4.2.1 Experimental Scope and Constraints

The evaluation focuses on proof-of-concept demonstrations: successful growth toward predefined targets and qualitative assessment of structural coherence. Quantitative metrics for

long-term stability, systematic regeneration under controlled damage patterns, or comparison with other 3D generative methods are limited.

The *Minecraft* environment also constrains the experimental setup. Voxel states are mapped to a discrete set of block types, the rendering pipeline is fixed, and interaction occurs through the game's built-in tools. While this provides a real-time testbed accessible to a broad audience, it limits geometric complexity, material expressiveness, and the precision of experimental measurements available for systematic investigation.

4.2.2 Relation to This Work

Sudhakaran et al. [21] provide the methodological basis for extending neural CA from planar grids to volumetric $3 \times 3 \times 3$ perception neighbourhoods. Strict locality, parameter sharing, and gradient-based optimisation form the conceptual foundation adopted here.

The framework developed in this work embeds the 3D neural CA within Blender instead of a game engine. Blender provides programmatic access to volumetric grids, mesh representations, rendering pipelines, and Python scripting through its built-in API. This gives precise control over lattice resolution, boundary conditions, material properties, and intermediate-state inspection.

The goal is not only to demonstrate 3D feasibility, but to provide a research-oriented framework for controlled investigation of stability, persistence, and regeneration in volumetric neural CA. Where the Minecraft work showed that 3D NCA can produce visually convincing results in a game environment, this work aims to provide the tools for systematic experimentation and analysis.

4.3 Open-Source and Applied Three-Dimensional Neural Cellular Automata

Beyond the work of Mordvintsev et al. [13] and the interactive extension by Sudhakaran et al. [21], several open-source projects have explored three-dimensional neural CA. This section reviews implementations whose design decisions are directly relevant to the framework developed in this work.

4.3.1 Standard 3D Reconstruction Baseline

The repository of Aadityaza [9] implements a direct extension of the Growing NCA paradigm to three-dimensional space. It uses a Jupyter Notebook environment and PyTorch to reconstruct voxel shapes from a single seed.

This project provides a clean verification of 3D perception operators, specifically the expansion of Sobel-type filters into tri-axial gradient kernels for the x, y, and z axes. The implementation confirms that the convolution-based perception approach used in 2D NCA applies directly to volumetric tensors without losing spatial locality. For this work, the project served as an early reference point for validating the feasibility of the 3D extension.

The implementation is focused on single-target reconstruction and does not include a training server, interactive visualisation, or modular architecture for experimenting with different configurations. These are gaps that the framework developed in this work addresses.

4.3.2 Voxel-Based Morphogenetic Experiments

Rossi and collaborators [20] explored voxel-based neural CA for three-dimensional morphogenesis. Their implementation extends the 2D growing NCA to volumetric grids with $3 \times 3 \times 3$ perception filters and hidden channel representations.

The system trains to approximate 3D voxel targets from simple seeds. Its architecture follows the 2D formulation closely: fixed perception filters, residual updates, shared parameters, and gradient-based optimisation. The experiments confirm that volumetric growth works under strict locality constraints. However, the evaluation is limited to demonstrating successful growth toward targets. Systematic analysis of long-term persistence, regeneration capacity, or scaling behaviour under increasing resolution is not provided.

Both open-source projects provide proof-of-concept validation for 3D morphogenetic NCA but leave open questions about stability and robustness in larger volumetric domains. They also lack integration with a 3D modelling environment that would allow researchers to interactively visualise and manipulate the NCA during training.

4.4 Chapter Conclusion

The works reviewed in this chapter established the architectural principles – local perception, shared neural update rules, residual dynamics, and gradient-based optimisation – that form the methodological basis for this work. The original 2D Growing NCA demonstrated that a small neural network with fewer than 10 000 parameters can learn to grow, maintain, and regenerate complex patterns through purely local interactions. Subsequent three-dimensional extensions confirmed that these principles transfer to volumetric grids and interactive environments, though systematic evaluation in 3D remains limited.

The framework developed in this work builds on these foundations. It integrates volumetric perception, shared update networks, and structured training regimes within a controlled 3D setting. Unlike prior implementations, it is designed to be modular and extensible, providing a general experimental environment where different architectures, training strategies, and evaluation procedures can be compared systematically. The integration with Blender provides interactive visualisation and mesh-based target specification, addressing the tooling limitations identified in existing open-source 3D NCA projects.

Chapter 5

Architectural Design

This chapter presents the design of the 3D NCA framework and its overall architecture. It introduces the key components that form the framework and describes how these components interact with each other. The aim is to provide a clear conceptual overview of the system and explain the roles of its individual parts. To maintain focus on design, implementation-specific details such as the libraries and frameworks chosen are intentionally omitted here and discussed separately in the Implementation chapter.

5.1 Design Goals and Requirements

The purpose of the framework is to provide an environment for experimenting with neural cellular automata operating on three-dimensional voxel grids. To support experimentation and future extensions, the architecture requires a clear structure and well-defined interactions between its components.

The primary design goal is to enable simulation and training of 3D neural cellular automata while maintaining a modular architecture. The system should allow individual components, such as the simulation grid, update mechanism, and training pipeline, to be developed and modified independently. This simplifies experimentation with different models and update rules.

A second objective was to keep the core functionality independent from visualisation tools and external applications. The framework is therefore designed as a standalone module that provides the core functionality required for initialising and training neural cellular automata. Other systems can build on this module and reuse its computational components.

During development, practical constraints related to visualisation and execution environments influenced the design. Combining interactive visualisation with computationally intensive training within a single environment proved difficult. These limitations motivated a separation between computation and visualisation.

As a result, the system evolved toward an architecture where the core framework provides the underlying simulation and training functionality, while additional components use this functionality to provide visualisation and interaction. This approach preserves the independence of the framework while allowing it to be integrated into different workflows.

5.2 System Overview

The framework is organised into three layers that work together to provide an interactive environment for training and visualising three-dimensional neural cellular automata. Figure 5.1 shows the high-level structure and the communication paths between these layers.

The innermost layer is the **core framework**. It contains the NCA model and the training pipeline and has no dependency on any server or graphical environment. It can be used directly from Python scripts or Jupyter notebooks, as the accompanying experiment notebooks demonstrate.

The second layer is the **server**, which exposes the core framework over a TCP connection. A running server accepts a client connection, receives training configuration and target data, delegates execution to the core training pipeline, and streams state snapshots after each epoch back to the client as training progresses.

The third layer is the **Blender add-on**, which acts as graphical interface and a client. It connects to the server over TCP, provides a panel-based interface for configuation and control over training sessions, and renders incoming state broadcasts as three-dimensional voxel meshes in the Blender viewport.

This separation serves two purposes. Training can run on a remote GPU machine or a Jupyter notebook while the user interacts with Blender on a local workstation, with a tunnelling service bridging the two. At the same time, the core framework remains independent of both the server and the graphical environment, so researchers who do not need interactive visualisation or want to use their own training algorithms can import and use it directly in their own scripts or notebooks.

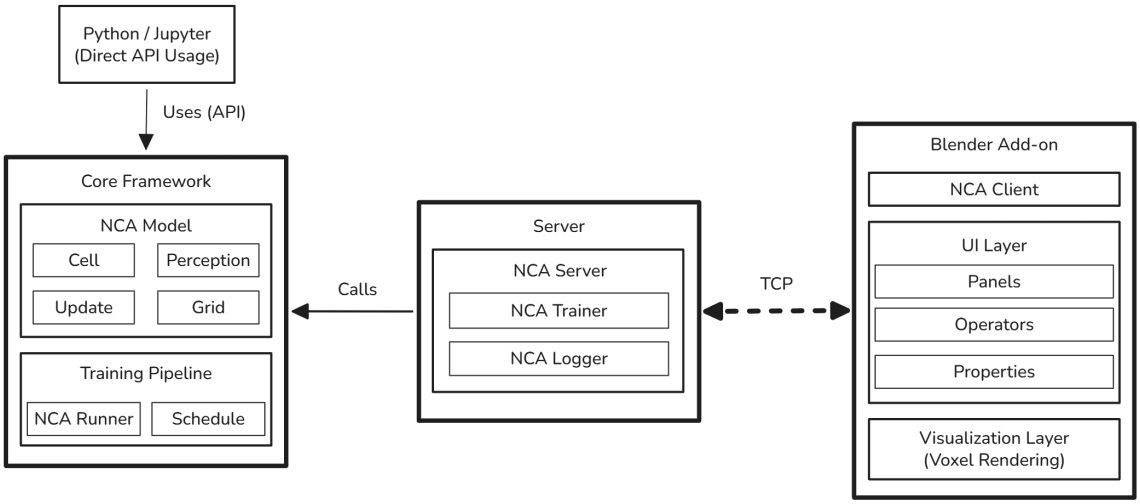

Figure 5.1: High-level system architecture. Arrows indicate dependency direction. The dashed TCP boundary marks the network separation between the server and the Blender client.

5.3 Core NCA Model

The core NCA model is composed of four components: `CellState`, `Perception3D`, `UpdateRule`, and `Grid3D`. Each component has its own configuration and can be instantiated indepen-

dently. `Grid3D` acts as the composition root, wiring the other three together and exposing a single-step forward pass. A higher-level wrapper, `NCAModel`, bundles all configurations into one object and delegates to `Grid3D` internally, providing a simpler entry point for common use cases.

Keeping configurations and components separate means that a researcher who wants to experiment with a custom update rule or a different perception kernel can replace just that component while leaving the rest of the pipeline unchanged. The experiment notebooks included with this work illustrate this directly: they use the model components without the runner or the server.

5.3.1 Cell

The cell component defines the structure of the per-voxel state vector and is responsible for computing the alive mask that gates cell updates. It is configured by specifying the number of hidden channels, the number of visible channels, and the alive threshold.

The state vector of each voxel contains three groups of channels, as shown in Figure 5.2. Hidden channels carry internal memory that is not directly supervised and is not compared against the target during training. Visible channels represent the quantities compared against the target, such as RGBA colour values or a single occupancy channel. An optional third group, task channels, allows a single model to be trained on multiple targets simultaneously using a one-hot encoding to distinguish between them. Task channels remain constant for the duration of a training phase, while hidden and visible channels evolve freely under the learned update rule.

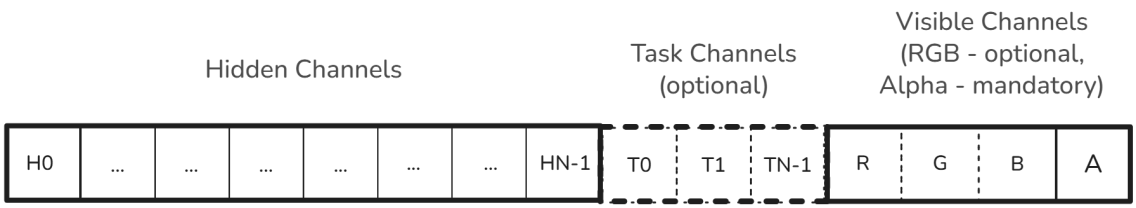

Figure 5.2: Per-voxel state vector layout. The vector contains hidden channels, optional task channels encoded as a one-hot vector, and visible channels. The alpha channel, which serves as the alive signal, is always the last visible channel.

The alive mask controls which regions of the grid are active. It is computed by pooling the alive channel over the local neighbourhood of each voxel and comparing the result against the configured threshold. A cell is considered potentially alive if either it or at least one of its immediate neighbours carries a value above the threshold. This neighbourhood pooling has an important consequence for growth: a dead cell adjacent to an alive cell is included by the mask, which allows it to receive nonzero updates and transition to an active state over time [13].

5.3.2 Perception

The perception component converts the current grid state into a per-cell representation by applying a fixed set of three-dimensional convolutional filters over the local neighbourhood.

The filters are non-learnable; their weights are fixed at initialisation and do not change during training. Three filter groups are applied across all channels. The identity filter

returns each cell's own current state unchanged, giving the update rule direct access to what the cell already contains. The neighbour sum filter aggregates the six face-adjacent neighbours, capturing the local occupancy and composition of the immediate neighbourhood. The Laplacian filter measures how much a cell's state differs from its neighbourhood average, encoding local spatial variation. Applying all three groups across all C channels produces a perception vector at each cell location with dimensionality $3 \times C$.

Keeping the filters fixed follows the original design of Mordvintsev et al. [13]. In practical terms, this means the perception stage has no trainable weights and cannot overfit on its own. Translation invariance is built in by construction through the filter layout, not learned implicitly from data. The only trainable part of this stage is therefore the update rule.

5.3.3 Update Rule

The update rule maps each cell's perception vector to a state increment. Its configuration controls the capacity of the underlying network and whether stochastic update masking is active.

As in Mordvintsev et al. [13], the increment is added residually instead of replacing the previous state:

$$\mathbf{s}_{t+1} = \mathbf{s}_t + \Delta\mathbf{s}.$$

This tends to produce gradual local adjustments rather than abrupt jumps, which is important when the model is unrolled for many simulation steps.

The same update network is applied independently at every cell. This preserves the locality constraint, because each increment depends only on local perception. It also keeps the dynamics spatially homogeneous: no cell has a privileged role by design. Any functional differences between cells must emerge from their local context in the grid.

When stochastic masking is enabled, each step updates only a random subset of cells. This helps avoid lockstep synchronisation artefacts and generally leads to dynamics that better tolerate asynchronous perturbations. In practice, that robustness becomes especially relevant in long-horizon persistence and regeneration experiments [13].

5.3.4 Grid and Model Composition

`Grid3D` composes the three components above into a module that executes one complete NCA update step. Figure 5.3 shows the data flow through a single step.

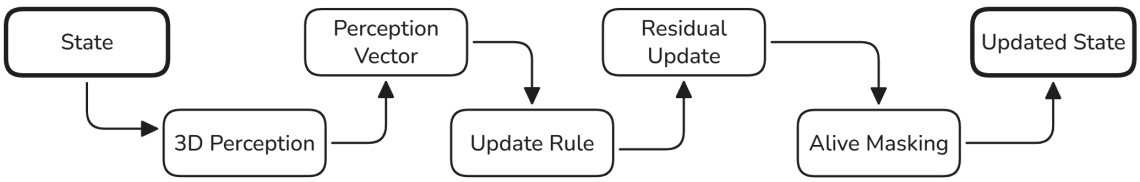

Figure 5.3: Data flow through a single NCA update step. The state passes through the fixed 3D perception filters, is transformed by the shared update rule, and is then gated by the alive masking mechanism before producing the updated state.

A single step proceeds as follows. The current state is passed to the perception component, which applies the fixed filters and produces a perception vector at every cell location. The update network produces a state increment, which is added residually to the current

state. A post-step alive mask is then computed from the updated state, and any cell that falls below the alive threshold has its state zeroed out.

At the architectural level, masking keeps activity close to the part of the structure that is developing and helps avoid random activation in empty regions. As a result, growth stays more controlled and the dynamics remain more stable over longer runs.

`NCAModel` is a higher-level entry point that bundles all component configurations and delegates to `Grid3D` internally. It additionally handles model persistence: a saved checkpoint stores the configuration alongside the learned weights, making the checkpoint fully self-describing and loadable for inference without access to the original training setup.

5.4 Training Pipeline

The training pipeline sits above the core NCA model and provides the infrastructure needed to optimise the model parameters toward a target voxel configuration. It consists of two components: a training runner, which implements the training loop, and a schedule manager, which handles dynamic parameter changes during training.

5.4.1 Runner

The runner implements pool-based training. Rather than restarting from a fresh seed on every iteration, it maintains a fixed-size pool of persistent states that represent the grid at various stages of development. At each iteration, a batch is sampled from the pool, evolved for a randomly chosen number of steps, evaluated against the target, and written back. The sample with the highest loss is replaced by a fresh seed, which maintains pool diversity and prevents the pool from converging to a single developmental stage [13].

The training objective has three weighted parts. Alpha loss compares the predicted and target alive channels. Colour loss compares RGB values only in areas where the target is present, so shape and colour are learned together. Overflow loss penalises activity outside the target, which helps limit unwanted growth. All three weights are configurable and can be changed at runtime through the schedule system.

The number of forward steps applied per training iteration increases over the course of training according to a linear curriculum. Early iterations use short unrolls, which keep gradient paths manageable and allow the model to learn coarse shape quickly. Later iterations use longer unrolls, which test whether the learned rule maintains the target configuration over extended simulation horizons. Both the minimum and maximum step counts increase linearly over a fixed number of epochs and then stabilise at their final values.

5.4.2 Schedule

The schedule component manages a list of one-shot events that fire at specified training epochs. Supported event types include changes to the learning rate, batch size, and individual loss weights, as well as target changes that replace the voxel configuration the model is training toward mid-session. Events can be added, removed, or replaced in bulk at any point during training.

The schedule is designed to be thread-safe, since the training loop and the server message handler may access it concurrently. Events can be configured to fire immediately on the next check rather than waiting for a specific epoch, which is useful when the user wants to apply a change without further delay during a live session.

5.5 Server and Networking

The server layer exposes the training pipeline over a TCP connection. It consists of three main components: a connection manager, which handles clients and dispatches messages; a session runner, which executes training and inference; and a logger, which persists run metadata and artefacts to disk. A shared protocol module defines the message format used across the boundary between the server and the Blender client.

5.5.1 Connection Architecture

5.6 Core Framework

TODO

5.7 NCA Model Components

TODO

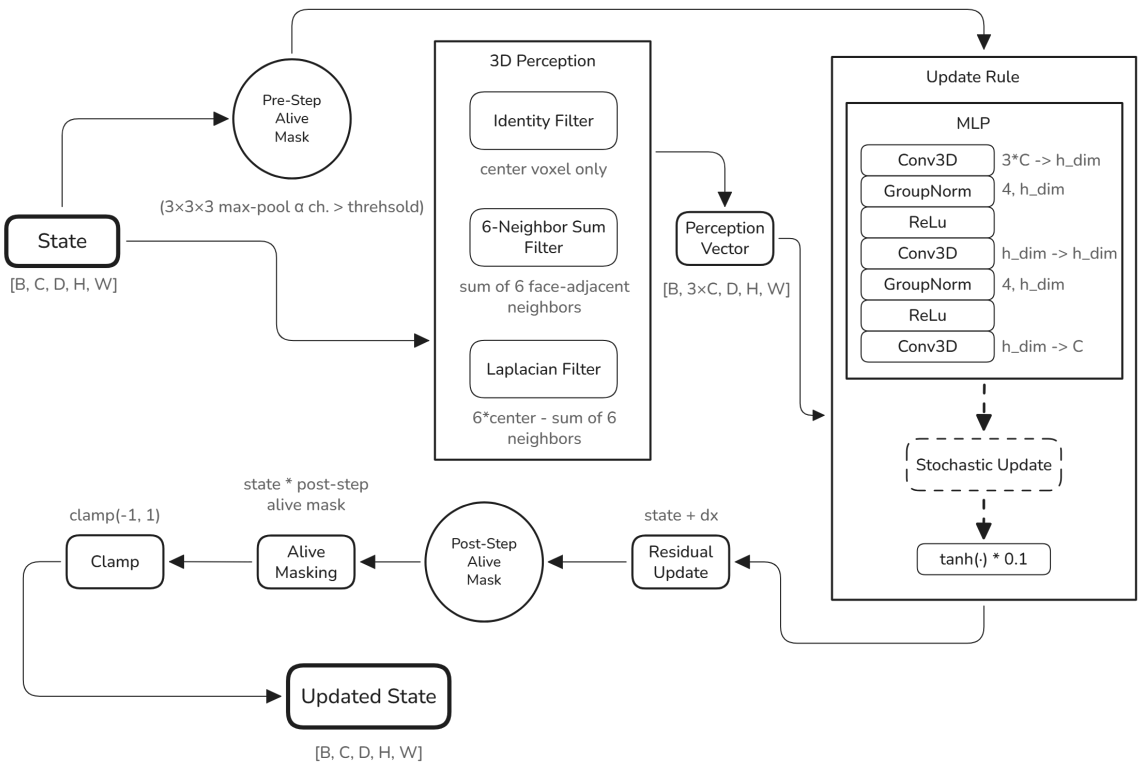

Figure 5.4: Detailed data flow through a single NCA update step, showing the full layer sequence within the update MLP, the optional stochastic fire mask, and the bounded tanh output.

5.8 Training Pipeline

TODO

5.9 Server

TODO

5.9.1 Protocol

TODO

	Header	Payload
Size	4 bytes	N bytes
Encoding	uint32, big-endian	UTF-8, JSON
Value	N (payload lenght)	JSON object (type + message-specific content)

Figure 5.5: Wire message format. Each message consists of a four-byte big-endian unsigned integer carrying the payload byte length, followed by the UTF-8 JSON payload.

Message Type	JSON Fields
init	type, config{}, target (base64), target_shape[]
run_model	type, model_b64, phase_steps, broadcast_every
update_schedule	type, events[]
stop	type
pause	type
resume	type
state	type, data (base64), shape[], epoch, loss
ack	type, message
error	type, message

Figure 5.6: Protocol message types and their JSON fields. Messages above the divider are sent by the client; messages below are sent by the server.

5.9.2 Logging

TODO

5.10 Blender Add-on

TODO

Chapter 6

Testing

TODO

Chapter 7

Experiments

TODO

Chapter 8

Conclusion

TODO

Bibliography

[1] Berlekamp, E. R.; Conway, J. H. and Guy, R. K. *Winning Ways for Your Mathematical Plays, Volume 2*. Academic Press, 1982.

[2] Chan, B. W.-C. *Lenia: Biology of Artificial Life*. 2019.

[3] Community, P. *PyTorch Documentation*. 2025. Available at: https://docs.pytorch.org/stable/index.html.

[4] Cook, M. *Universality in Elementary Cellular Automata*. 2004.

[5] Gardner, M. *Mathematical Games: The fantastic combinations of John Conway's new solitaire game "Life"*. 1970. Available at: https://web.stanford.edu/class/sts145/Library/life.pdf.

[6] Goodfellow, I.; Bengio, Y. and Courville, A. *Deep Learning*. MIT Press, 2016. Available at: https://www.deeplearningbook.org.

[7] Horváth, D. and Levin, M. *Self-Classifying MNCA Distill*. 2021. Available at: https://distill.pub/2021/self-classifying-mnca/.

[8] Ilachinski, A. *Cellular Automata: A Discrete Universe*. World Scientific, 2001. ISBN 981-02-4623-4.

[9] Jha, A. *3D Growing Neural Cellular Automata*. GitHub, 2024. Available at: https://github.com/Aadityaza/3d-Growing-neural-cellular-automata. Originally published under username Aadityaza.

[10] Kinderen, E. de and Wuensche, A. 3D CA in a fluid dynamics PDE. In: *Proceedings of the Second International Conference on Evolvable Systems: From Biology to Hardware*. Springer, 1998.

[11] Laine, S. and Karras, T. Efficient sparse voxel octrees. In: *Proceedings of the 2010 ACM SIGGRAPH Symposium on Interactive 3D Graphics and Games*. New York, NY, USA: Association for Computing Machinery, 2010. ISBN 9781605589398. Available at: https://doi.org/10.1145/1730804.1730814.

[12] Marinescu, D. C. Chapter 2 - Nature-Inspired Algorithms and Systems. In: Marinescu, D. C., ed. *Complex Systems and Clouds*. Boston: Elsevier, 2017, p. 33–63. Computer Science Reviews and Trends. ISBN 978-0-12-804041-6. Available at: https://www.sciencedirect.com/science/article/pii/B9780128040416000025.

[13] Mordvintsev, A.; Randazzo, E.; Niklasson, E. and Levin, M. *Growing Neural Cellular Automata*. 2020. Available at: https://distill.pub/2020/growing-ca/.

[14] Museth, K. VDB: High-resolution sparse volumes with dynamic topology. *ACM Trans. Graph.* New York, NY, USA: Association for Computing Machinery, 2013. ISSN 0730-0301. Available at: https://doi.org/10.1145/2487228.2487235.

[15] Neumann, J. von. *Theory of Self-Reproducing Automata.* University of Illinois Press, 1966.

[16] Niklasson, E.; Mordvintsev, A.; Randazzo, E. and Levin, M. Self-Organising Textures. *Distill*, 2021. Https://distill.pub/selforg/2021/textures/.

[17] Paszke, A.; Gross, S.; Massa, F.; Lerer, A.; Bradbury, J. et al. *PyTorch: An Imperative Style, High-Performance Deep Learning Library.* 2019. Available at: https://arxiv.org/abs/1912.01703.

[18] Rafler, S. *Generalization of Conway's "Game of Life" to a continuous domain – SmoothLife* arXiv preprint arXiv:1111.1567. 2011. Available at: https://doi.org/10.48550/arXiv.1111.1567.

[19] Randazzo, E.; Mordvintsev, A. and Levin, M. *Persistent Memory in Growing Neural Cellular Automata* Google Research Blog. 2021. Available at: https://ai.googleblog.com/2021/12/persistent-memory-in-neural-cellular.html.

[20] Rossi, F. and Orsini, M. *3D Growing Neural Cellular Automaton.* GitHub, 2021. Available at: https://github.com/SkyLionx/3d-cellular-automaton. Originally published under username SkyLionx.

[21] Sudhakaran, S.; Grbic, D.; Li, S.; Katona, A.; Najarro, E. et al. *Growing 3D Artefacts and Functional Machines with Neural Cellular Automata.* July 2021, ALIFE 2021: The 2021 Conference on Artificial Life, p. 108. ALIFE 2022: The 2022 Conference on Artificial Life. Available at: https://doi.org/10.1162/isal_a_00451.

[22] Tobias, P. A. G. *3D Cellular Automata.* 2009. Available at: https://eprints.bournemouth.ac.uk/17408/.

[23] Turing, A. M. The Chemical Basis of Morphogenesis. *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, 1952, vol. 237, no. 641, p. 37–72.

[24] Wolfram, S. *Statistical Mechanics of Cellular Automata.* 1983.

[25] Wolfram, S. *A New Kind of Science.* Wolfram Media, 2002. ISBN 1-57955-008-8. Available at: https://www.wolframscience.com/nks/.